Detecting *Sargassum* Bloom Directly From Satellite Top-of-Atmosphere Reflectance With High-Resolution Images

Hailong Zhang, Jianqiang Liu, Xiaomin Ye, Yan Bai, Deyong Sun, Shengqiang Wang, and Jingming Dong

***Abstract*— Massive floating *Sargassum* blooms have occurred frequently in many parts of the global ocean, and satellite remote sensing provides an effective way to monitor their spatiotemporal variation. Coarse-resolution satellite data, however, often suffer from a data gap in nearshore waters and detection uncertainty of *Sargassum* amount, especially for smaller patches. These limitations may be ameliorated by high-resolution satellite data, yet such great potential is hindered by the lack of reliable and easy-to-implement methods to detect *Sargassum* slicks. Here, combining the visible and near-infrared (NIR) top-of-atmosphere reflectance (R_{TOA}) data with the random forest (RF) model, a new method, namely the R_{TOA}-RF model, was designed to automatically detect *Sargassum* from high-resolution satellite imagery with four wavebands. Specifically, this model was successfully applied to various satellite sensors, including Gaofen1-wide field view multispectral camera (GF1-WFV; 16 m), GF2 multispectral scanner (GF2-MSS; 4 m), GF6-WFV (16 m), Huanjing1A/B charge-coupled device (HJ1A/B-CCD; 30 m), Haiyang1C/D coastal zone imager (HY1C/D-CZI; 50 m). Comparisons with visual inspection and cross-index indicated that all achieved satisfactory performance for detecting *Sargassum* slicks, with overall accuracy (OA) and *Kappa* values greater than 96% and 88%, respectively. The sensitivity analysis of the model as an example of different sensors suggested that the R_{TOA}-RF model can effectively identify *Sargassum* features under complex ocean background, cloud cover, and sunglint, even under conditions of surface wave glitter and weak *Sargassum* feature with low false positive rate (<9.60%). The findings here not only pave the way for operational monitoring of fine-scale *Sargassum*, but importantly provide thought for developing detection method of macroalgae blooms in global water.**

***Index Terms*— High-resolution satellite image, macroalgae detection, random forest (RF) model, *Sargassum*, top-of-atmosphere reflectance (R_{TOA}).**

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Hailong Zhang is with the School of Marine Sciences, Nanjing University of Information Science and Technology, Nanjing 210044, China, also with the Key Laboratory of Space Ocean Remote Sensing and Application, MNR, Beijing 100081, China, and also with the State Key Laboratory of Satellite Ocean Environment Dynamics, MNR, Hangzhou 310012, China (e-mail: zhanghl@nuist.edu.cn).

Jianqiang Liu and Xiaomin Ye are with the National Satellite Ocean Application Service, Beijing 100081, China, and also with the Key Laboratory of Space Ocean Remote Sensing and Application, MNR, Beijing 100081, China (e-mail: jqliu@mail.nsoas.org.cn; yexiaomin2011@foxmail.com).

Yan Bai is with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, MNR, Hangzhou 310012, China (e-mail: baiyan@sio.org.cn).

Deyong Sun and Shengqiang Wang are with the School of Marine Sciences, Nanjing University of Information Science and Technology, Nanjing 210044, China, and also with the Key Laboratory of Space Ocean Remote Sensing and Application, MNR, Beijing 100081, China (e-mail: sundeyong@nuist.edu.cn; wsqiang815@163.com).

Jingming Dong is with Lianyungang Meteorological Bureau, Lianyungang 222000, China (e-mail: djm331487563@163.com).

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I. Introduction

In RECENT years, marine macroalgae blooms of floating *Sargassum* (often called golden tide) have been often reported around the world and getting increasingly more attention [1], [2], [3], [4]. Pelagic *Sargassum* across the Atlantic Ocean has increased significantly during the past decade [5], [6]. In the Yellow and East China Seas (YECS), the prevalence of *Sargassum* bloom has become a local ecological disaster [3], [7]. Massive *Sargassum* blooms cause serious local economic and ecological problems. Timely and accurate monitoring and tracking of *Sargassum* blooms is crucial for decision-making in disaster management, thus mitigating the negative effects. Satellite remote sensing is an effective tool to provide better *Sargassum* information, such as distribution, areal coverage, and drifting trajectory, because of its own strengths in synoptic and frequent observations [8].

Optical imagery from various satellite sensors has been used for monitoring *Sargassum* blooms and follow-up researches [9], [10], [11], [12]. To date, coarse-resolution satellite images have been commonly and successfully used to study *Sargassum* events around the world, including moderate resolution imaging spectroradiometer (MODIS), medium resolution imaging spectrometer (MERIS), and geostationary ocean color imager (GOCI). The wide-range scene measurements of coarse-resolution satellite sensors show some advantages in detecting large-scale *Sargassum* blooms. Each sensor,

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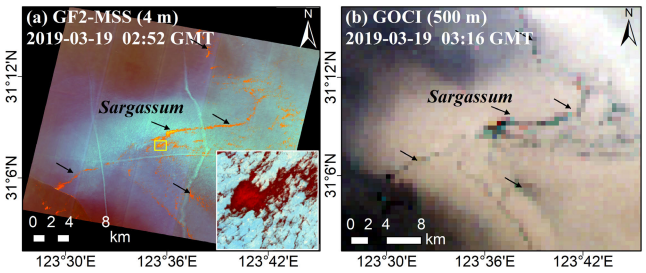


Fig. 1. *Sargassum* slicks in (a) high-resolution GF2-MSS image (4 m spatial resolution) and (b) coarse-resolution GOCI image (500 m spatial resolution).

however, has its own lower detection limit, such as 0.2% of a pixel size for 1 km MODIS image [13]. In this case, small *Sargassum* patches may "loss" in coarse-resolution images (see Fig. 1 for example), particularly in the early-stage of macroalgae outbreak. In addition, the data quality of coarse-resolution image pixels is compromised in coastal waters, mainly caused by the interference of land adjacency effects, shallow-water bottom, and higher turbid waters [14]. For example, there are no valid MODIS observations in nearshore waters [15]. These disadvantages increase the difficulty in tracing the origin of the *Sargassum* bloom and hinder management actions. Fortunately, the limitations may be overcome by satellite imagery with high resolution. One advantage of high-resolution images is the ability to detect the small-scale macroalgae features, and thereby, more accurately evaluate their abundance. For instance, *Sargassum* slicks can be clearly visualized in the GaoFen2 multispectral scanner (GF2-MSS) image with 4 m spatial resolution shown in Fig. 1(a), while they were not available from the coarse-resolution 500 m GOCI image shown in Fig. 1(b). Although many high-resolution satellite sensors were designed primarily for land-based applications, some measurements were already used for studying the green tide of *Ulva prolifera* (*Ulva*) [8], [16], [17]; however, they were rarely used in *Sargassum* monitoring. Most high-resolution satellite sensors, specifically within about 30 m spatial resolution, are equipped with only four wavebands, i.e., blue, green, red, and near-infrared (NIR) bands. With the limited spectral information, it is important to find an effective approach to capture *Sargassum* slicks from high-resolution imagery.

Naturally, several optical remote sensing methods have been proposed to detect *Sargassum*, and all were dependent on the distinct spectral properties between *Sargassum* and non-*Sargassum*. They can be mainly divided into two categories, namely "threshold-based" and "machine learning" methods. The "threshold-based" methods employ various algebraic forms of the multiband spectrum to elevate the *Sargassum* feature and then choose the appropriate threshold value to detect *Sargassum*pixels. For example, the maximum chlorophyll index (MCI) [1] with the use of MERIS radiance data at 681, 709, and 753 nm was applied to *Sargassum* monitoring in the North Atlantic and Gulf of Mexico. Similarly, using the Rayleigh-corrected reflectance (R_{rc}) data at red, NIR, and shortwave infrared (SWIR) bands, the floating algae index (FAI) [18] and alternative FAI index [19] were designed for *Sargassum* detection. These algorithms, however, cannot be applied to high-resolution satellite sensors due to the lack of the required wavebands. Based on Chinese GF1-wide field view multispectral camera (WFV) images, the difference vegetation index (DVI) defined as the surface reflectance (*Ref*) difference of NIR and red wavebands was designed to detect *Sargassum* in the Yellow Sea [20], but may increase the uncertainty in detecting weak *Sargassum* slicks. Additionally, for all threshold-based methods, threshold selection is inevitable and dependent on method performance. In practice, the optimal threshold value may be adjusted for individual satellite images due to variable image backgrounds.

Instead, the "machine learning" methods for target classification can bypass the step of threshold selection. Several machine learning techniques were already applied in satellite imagery to identify floating macroalgae of *Sargassum* or *Ulva* [21]. Using GOCI R_{rc} data with eight wavebands, the multilayer perceptron was applied to monitor floating macroalgae blooms in the YECS [22], but its feasibility and effectiveness in high-resolution satellite imagery with four wavebands need to be further verified. Based on satellite R_{rc} data, a VGGUnet-based framework developed by Wang and Hu [15] and a two-step random forest (RF) approach designed by Xiao et al. [23] were respectively used toautomatically extract *Sargassum* features from high-resolution satellite images. The design of these methods was based on both satellite R_{rc} or *Ref* signals. Although Rayleigh scatting contributions can be calculated accurately for most observation geometries, the error will exceed 10% under high solar zenith angle condition [24] and even make the image useless under significant sunglint [25] that often occur in a high-resolution satellite image. Furthermore, the implementations of Rayleigh scattering and atmospheric correction processes to generate R_{rc} and *Ref* data need additional information and more running time [8]. Comparatively, satellite top-of-atmosphere reflectance (R_{TOA}) data are easier to obtain from satellite digital number (DN) data only via simple algebra. Zhang et al. [17] clarified the practicability of satellite R_{TOA} signal on macroalgae *Ulva* detection, and did develop an effective R_{TOA}-based detection algorithm with a high-resolution satellite image. The use of satellite R_{TOA} signal should have the potential to detect floating *Sargassum* bloom.

The objective of this work, therefore, was to evaluate the feasibility of satellite R_{TOA} to extract the *Sargassum* feature, and thus, develop an R_{TOA}-based automatic method with an RF model (namely R_{TOA}-RF model) for detecting *Sargassum* blooms from multisensor high-resolution satellite images. This article was structured as follows: multisensor satellite data and processing were first described, followed by the details of the proposed R_{TOA}-RF model for *Sargassum* detection. We evaluated its performances on various high-resolution satellite images, including GF1-WFV, GF2-MSS, GF6-WFV, Huanjing1A/B charge-coupled device (HJ1A/B-CCD), Haiyang1C/D coastal zone imager (HY1C/D-CZI). The model sensitivity to the observing conditions was then analyzed. Last, a comparison of the model performances with different input signals (R_{TOA}, R_{rc}, and *Ref*) and the strengths and limitations of the R_{TOA}-RF model were discussed.

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TABLE I
INFORMATION OF HIGH-RESOLUTION SATELLITE SENSORS USED IN THIS STUDY

Parameter		GF1-WFV	GF2-MSS	GF6-WFV	HJ1A/B-CCD	HY1C/D-CZI
Band setting (nm)	blue	450-520	450-520	450-520	430-520	420-520
	green	520-590	520-590	520-590	520-600	520-600
	red	630-690	630-690	630-690	630-690	610-690
	NIR	770-890	770-890	760-900	760-900	760-890
Spatial resolution		16 m	4 m	16 m	30 m	50 m
Revisit time		2 days	5 days	2 days	2 days	3 days

II. DATA AND METHODS

A. Satellite Data and Preparations

1) GF Satellite Constellation: The Chinese GF constellation launched in succession in 2013 can provide multisensor optical satellite imagery with high spatial-temporal resolution, and thus, they are a good dataset choice for studying *Sargassum*. Here the GF1-WFV, GF6-WFV, and GF2-MSS sensors were selected for demonstration purposes. GF1/6-WFV and GF2-MSS sensors have four wavebands (i.e., blue, green, red, and NIR), and their details are summarized in Table I. In this study, satellite DN images of 23 GF1-WFV, ten GF2-MSS, and two GF6-WFV collected in the YECS during 2015–2021 were downland from the China Center for Resource Satellite Data and Application (CRESDA; http://www.cresda.com/CN). In addition, one GF1-WFV image collected in the Gulf of Mexico was also obtained to confirm the extension of the method application. The satellite DN images were first processed to radiance L_λ and then generated to satellite $R_{TOA,\lambda}$ data using the following equations:

$$L_\lambda = \mathrm{DN}_\lambda \cdot \mathrm{gain}_\lambda + \mathrm{bias}_\lambda \tag{1}$$

$$R_{TOA,\lambda} = \pi d^2 \cdot L_\lambda / (E\mathrm{sun}_\lambda \cdot \cos\theta_s) \tag{2}$$

where gain_λ, bias_λ, and $E\mathrm{sun}_\lambda$ are the gain, bias, and mean exo-atmospheric solar irradiance values at waveband λ for a given sensor, respectively; θ_s and d are the solar zenith angle in degree and the earth-to-sun distance in astronomical units for a given satellite image, respectively. In addition, the GF1-WFV2 surface reflectance (*Ref*) data collected on 18 May 2017 was produced using FLAASH atmospheric correction model, and then satellite virtual-baseline floating macroalgae height (VBFAH [16]) data were obtained following (3). Here the VBFAH-detected *Sargassum* result was compared with that of the R_{TOA}-RF model to verify its accuracy

$$\mathrm{VBFAH} = \left(Ref_{NIR} - Ref_{green}\right) + \frac{\left(Ref_{green} - Ref_{red}\right) \cdot \left(\lambda_{NIR} - \lambda_{green}\right)}{2\lambda_{NIR} - \lambda_{red} - \lambda_{green}} \tag{3}$$

where $\lambda_{green} = 557$ nm, $\lambda_{red} = 676$ nm, $\lambda_{NIR} = 774$ nm for GF1-WFV2 sensor.

2) HJ1A/B-CCD and HY1C/D-CZI Data: To study the applicability of the proposed model design, we selected the other two high-resolution satellite sensors: HJ1A/B-CCD and HY1C/D-CZI, with 30 and 50 m spatial resolution, respectively. Their spectral bands are similar to GF-WFV sensors, as shown in Table I. In this study, thirteen HJ1A/B-CCD DN images over the YECS were downloaded from the CRESDA website and processed to satellite R_{TOA} data used for *Sargassum* extraction following (1) and (2). Meanwhile, eleven HY1C/D-CZI radiance images containing *Sargassum* blooms in the YECS were obtained from the National Satellite Ocean Application Service (NSOAS, http://www.nsoas.org.cn). HY1C/D-CZI R_{TOA} data were generated from satellite radiance data following (2).

3) Landsat8-OLI Data: The Landsat8-operational land imager (LC08-OLI) image collected on 16 March 2019 was selected to assess the model accuracy through cross-index comparison with simultaneous and colocated GF6-WFV image. The LC08-OLI Level-1 DN image was acquired from the U.S. Geological Survey (USGS) (https://www.usgs.gov) and processed to generate *Ref* data using FLAASH atmospheric correction model. The LC08-OLI FAI data were then obtained from *Ref* data, as shown in the following equation, and satellite *Sargassum* pixels were then identified using the FAI threshold approach [18]:

$$\mathrm{FAI} = \left(Ref_{NIR} - Ref_{red}\right) + \left(Ref_{red} - Ref_{SWIR}\right) \cdot \left(\lambda_{NIR} - \lambda_{red}\right) / \left(\lambda_{SWIR} - \lambda_{red}\right) \tag{4}$$

where $\lambda_{red} = 655$ nm, $\lambda_{NIR} = 865$ nm, $\lambda_{SWIR} = 1609$ nm for LC08-OLI sensor.

B. Sargassum Detection Method Using Satellite R_{TOA}

In this study, *Sargassum* detection from high-resolution satellite imagery followed a simple and straightforward workflow. Satellite R_{TOA} data were first generated from raw satellite images for a given sensor, as mentioned above. Then, the model input features of the proposed model were calculated (see Table II) and then used to train and validate the sensor-specific R_{TOA}-RF model. Finally, satellite *Sargassum* pixels were detected using the sensor-specific R_{TOA}-RF model. It should be noted that cloud class was one output of the R_{TOA}-RF model, and thus cloud masking was not considered.

1) Input Features of the R_{TOA}-RF Model: The model input features are the important parameters for the RF model [26], and the optimal input combination can significantly improve the final classification accuracy. In this study, through the detailed analysis of R_{TOA} spectral of the concerned classes, including *Sargassum*, seawater and cloud (see Section III-A), a total of nine input features were finally built for the R_{TOA}-RF model, including R_{TOA} data at four wavebands and five spectral indexes (i.e., NDWI, NDVI, SNR, Br, and Wh). The detailed information and calculation formulas of the nine model input features are listed in Table II, and here λ_{NIR} and λ_{red} are the sensor-specific center wavelengths of NIR and red bands, respectively. Once satellite R_{TOA} data are available, the model input features are easy to obtain.

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TABLE II
INFORMATION OF THE MODEL INPUT FEATURES OF THE R_{TOA}-RF MODEL

Abbreviation	Description	calculation formulae
$R_{TOA,\lambda i}$	R_{TOA} value at waveband λ_i; λ_i =blue, green, red, and NIR band, respectively	
NDWI	Normalized Difference Water Index	$(R_{TOA,green} - R_{TOA,NIR}) / (R_{TOA,green} + R_{TOA,NIR})$
NDVI	Normalized Difference Vegetation Index	$(R_{TOA,NIR} - R_{TOA,red}) / (R_{TOA,NIR} + R_{TOA,red})$
SNR	Spectral Slope of NIR and red wavebands	$(R_{TOA,NIR} - R_{TOA,red}) / (\lambda_{NIR} - \lambda_{red})$
Br	Brightness Index	$\sum_{i=1}^{4} R_{TOA,\lambda_i} / 4$
Wh	Whiteness Index	$\sum_{i=1}^{4} \left\| \left(R_{TOA,\lambda_i} - Br\right) / Br \right\|$

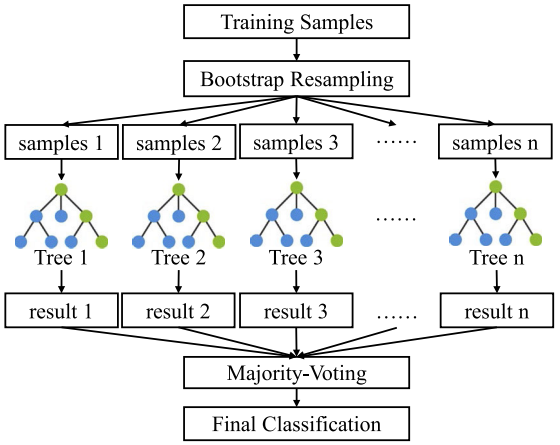


Fig. 2. Processing diagram of RF model.

2) R_{TOA}-RF Model for Sargassum Detection: The RF model was proposed by Breiman [26] based on the statistical learning theory and is an ensemble learning approach as an important branch of machine learning. It was commonly applied in object-oriented classification with remote sensing images [27], [28], [29] because of its high processing speed and classification accuracy, and insensitivity to overfitting. In the RF model, a set of classification and regression trees (CARTs) is used for prediction making. The training subsets called "in-bag samples" are generated from the original training dataset using the bootstrap algorithm and train each of the CARTs [30]. The remaining samples called "out-of-bag samples," are used to check model applicability. In brief, the process of RF model includes the following steps. Several sample subsets are randomly selected from original training samples using the bootstrap resampling approach, and each subset generates a decision tree and obtains prediction results. The final classification result is determined by combining the prediction results of all classification trees (Fig. 2). The number of classification trees (N_{Tree}) and the model input features are set for the R_{TOA}-RF model. In theory, the more classification trees, the more time it takes. Considering the computational efficiency, N_{Tree} was set experimentally to 50 according to the accuracy analysis of each additional five trees. As mentioned above, the nine model input features shown in Table II were selected for the R_{TOA}-RF model.

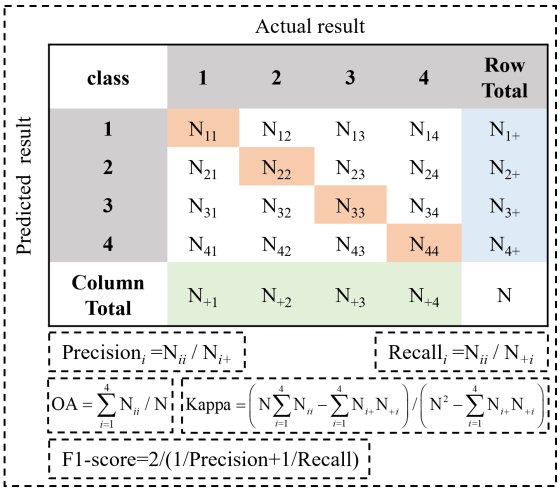


Fig. 3. Generic form of an error matrix and the calculation formulas of five statistical indicators.

C. Accuracy Evaluation

The error matrix, also known confusion matrix, was used in this study to evaluate the model performance. The error matrix is a common way of accuracy assessment for multiclass classification [31]. Five statistical indicators for the error matrix were computed, including precision, recall, $F1$-score, overall accuracy (OA), and *Kappa* coefficient [32]. Their calculation formulas and the generic form of the error matrix in this study are shown in Fig. 3. Here, classes 1–4 represent *Sargassum* I, *Sargassum* II, seawater, and cloud, respectively. Precision tells the class-specific ratio of the correctly classified pixels to the identified pixels, and recall tells the class-specific probability of the labeled pixels being classified correctly. $F1$-score is the harmonic mean of precision and recall. OA tells the proportion of the pixels correctly classified by the model in the total pixels. *Kappa* represents the proportion of error reduction between classification and completely random classification. OA and *Kappa* are most widely used in classification accuracy assessment.

When assessing the model performance on satellite images, three verification ways were adopted: 1) a qualitative comparison of the *Sargassum* result detected by the proposed model with visual interpretation; 2) the error matrix comparison of the predicted result and "true" result by manual identification for the selecting sample points; and 3) cross-comparison between the proposed method and existing indexes (VBFAH and FAI) using the synchronous image pairs of different sensors.

III. RESULTS

A. Spectral R_{TOA} of Satellite Sensors

As mentioned in Section II-B, the analysis of the spectral difference between *Sargassum* and non-*Sargassum* is important to build input features for the R_{TOA}-RF model, and thereby, contributes to the method's accuracy. Can the difference in the R_{TOA} spectral properties be clearly captured in high-resolution satellite images. Here, to better understand R_{TOA} spectral features for different sensors, extensive pixel

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TABLE III
NUMBER OF THE COLLECTED PIXEL SAMPLES FOR FIVE SATELLITE SENSORS

Sensor	*Sargassum* I	*Sargassum* II	Seawater	Cloud
GF1-WFV	130215	23590	237146	21258
GF2-MSS	114942	23542	137031	30480
GF6-WFV	8227	3894	13227	12484
HJ1A/B-CCD	25591	1175	42479	9088
HY1C/D-CZI	15017	4404	41732	15063

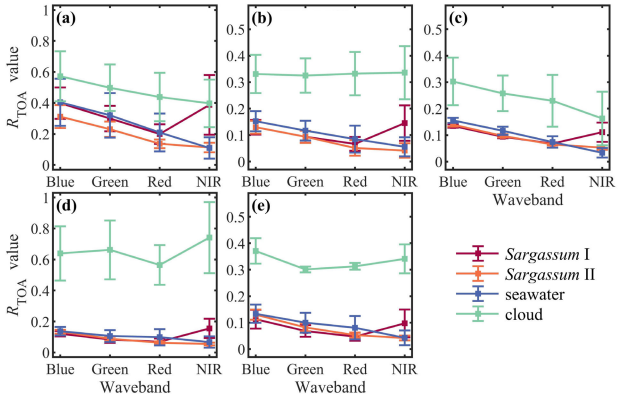


Fig. 4. Satellite R_{TOA} spectrum of four classes for different sensors. (a) GF1-WFV. (b) GF2-MSS. (c) GF6-WFV. (d) HJ1A/B-CCD. (e) HY1C/D-CZ1.

samples for the concerned targets (*Sargassum*, seawater, cloud) were randomly collected from several satellite images by artificial selection (see Table III). For example, extensive pixel samples (153 805 *Sargassum* pixels, 237 146 seawater pixels, and 21 258 cloud pixels) were collected from 23 GF1-WFV images.

Fig. 4 shows the satellite R_{TOA} spectrum of the concerned targets for various sensors used in this study. According to the spectral shape analysis of *Sargassum*, two typical shapes were found: 1) *Sargassum* I with $R_{TOA,NIR} > R_{TOA,red}$ (see crimson lines of Fig. 4) that occupy the majority of blooms and 2) *Sargassum* II with $R_{TOA,NIR} \leq R_{TOA,red}$ (see orange lines of Fig. 4), which mainly include weak *Sargassum*. The most significant spectral difference between *Sargassum* I and other classes was located in red and NIR wavebands. *Sargassum* II and seawater presented similar spectral shapes that R_{TOA} value decreased with the increase of wavebands from blue to NIR, but with differences in the red-NIR spectral slope (Fig. 4). This dilemma made it more difficult to distinguish them, as reported in Wang and Hu [14]. Here *Sargassum* II was set as a separate class in the R_{TOA}-RF model to improve classification accuracy. Satellite R_{TOA} values of cloud at visible and NIR wavebands were significantly higher than those of other classes. Thus, the distinct R_{TOA} spectral features of the four classes provided the basis for *Sargassum* detection using satellite R_{TOA} signal.

B. Sargassum Detection Using the R_{TOA}-RF Model From High-Resolution Satellite Data

According to the above analysis, five R_{TOA}-based spectral indexes (NDWI, SNR, NDVI, Br, and Wh), except for original R_{TOA} data at four wavebands shown in Table II, were specially

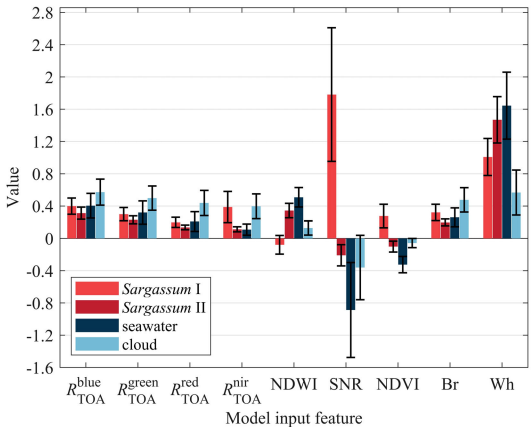


Fig. 5. Statistical results of the nine model input features for four target samples for the GF1-WFV sensor.

built as input to the proposed model. For GF1-WFV sensor as an illustrative example, Fig. 5 shows the statistical results (mean ± one standard deviation) of nine model input features for *Sargassum* I and II, seawater, and cloud. It was found that the NDWI values of seawater were higher than those of other classes, suggesting that it can be used to distinguish seawater and other classes. For SNR and NDVI features, the values of *Sargassum* I and II were higher than seawater and cloud, which means that the two features can enhance the *Sargassum* signal. It was observed that the value ranges of Br and Wh of clouds were different from those of *Sargassum* and seawater, implying that they could be used to identify clouds from other classes.

The sample datasets with class labels for any given satellite sensor shown in Table III were divided into training samples (70% of all samples) and testing samples (30% of all samples) to train and validate the R_{TOA}-RF model. For a given satellite sensor, the nine R_{TOA}-based model features were generated, and the specific-sensor R_{TOA}-RF model was effectively optimized using the corresponding training dataset. For the samples in the testing dataset, their detection results were then compared with the "true" results to calculate the corresponding statistical indicators. The classification results obtained showed promising performances of the R_{TOA}-RF model, with the OA, $F1$-score, and *Kappa* values greater than 95.00%.

To investigate the model performances on various satellite sensors, the specific-sensor R_{TOA}-RF models were applied on a separate group of representative GF1-WFV, GF2-MSS, GF6-WFV, HJ1A/B-CCD, and HY1C/D-CZI (Fig. 6). It was clearly found from Fig. 6 that the *Sargassum* pixels were mostly detected (i.e., low false negatives) using the R_{TOA}-RF model (middle panel), and the detected *Sargassum* slicks well matched the corresponding slicks in the RGB images (left panel). From visual inspection, no apparent noise signals were misidentified as *Sargassum* class (i.e., low false positives). The high agreement between *Sargassum* detection and visual identification were distinctly observed in the selected small regions (right panel of Fig. 6). Clearly, the satisfied performances of the R_{TOA}-RF model on various satellite images were achieved, suggesting that this model can accurately detect

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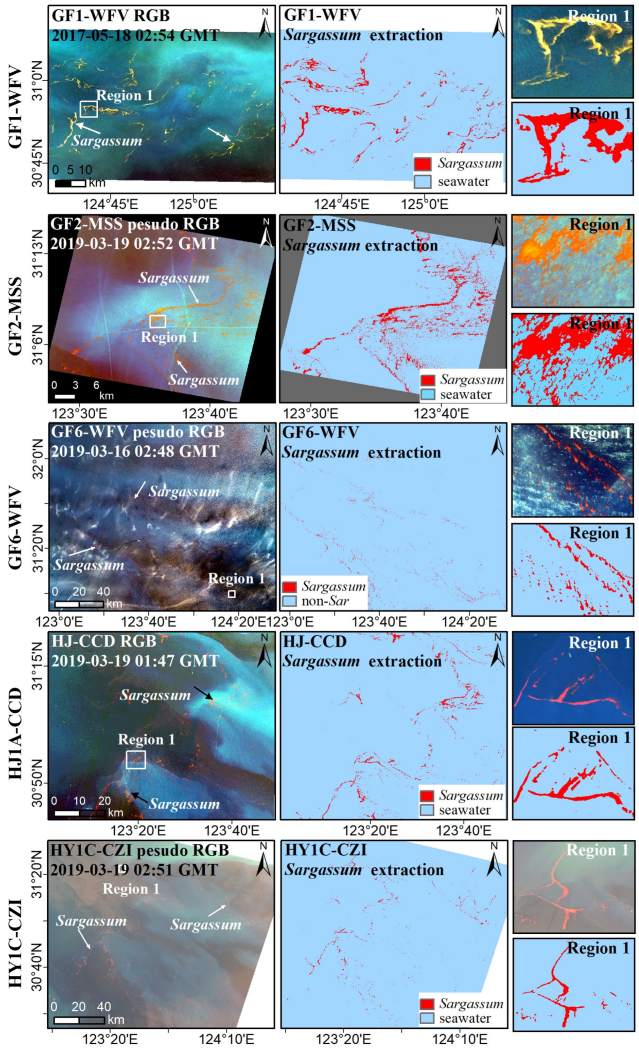


Fig. 6. *Sargassum* detection from multisensor high-resolution satellite images using the R_{TOA}-RF model. (Left panel) RGB images. (Middle panel) *Sargassum*detection results. (Right panel) More details about *Sargassum* extraction performance were displayed in the selected small regions.

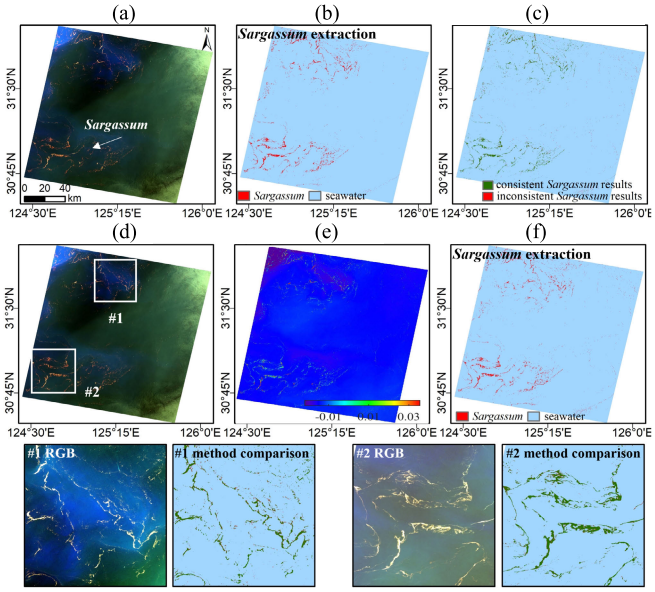


Fig. 7. Comparison of the *Sargassum* result detected by the R_{TOA}-RF model with the VBFAH-detected result using the GF1-WFV image on 18 May 2017. (a) GF1-WFV RGB. (b) R_{TOA}-RF model. (c) R_{TOA}-RF versus VBFAH method. (d) GF1-WFV RGB. (e) *Ref*-based VBFAH. (f) VBFAH index.

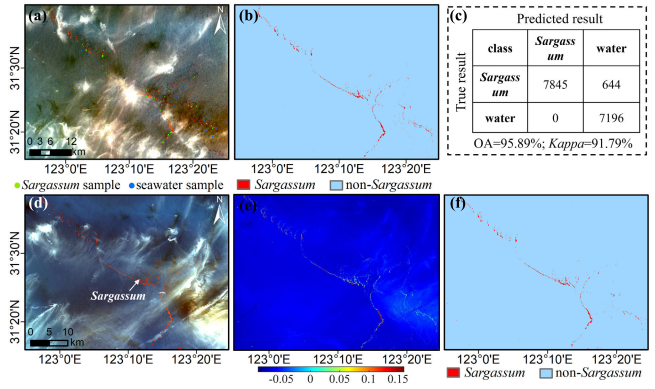


Fig. 8. Comparison of the GF6-WFV *Sargassum* detected by the R_{TOA}-RF model with the true result by manual identification and the LC08-OLI FAI-detected *Sargassum* result. (a) GF6-WFV RGB. (b) R_{TOA}-RF model. (c) Error matrix. (d) LC08-OLI RGB. (e) *Ref*-based FAI. (f) FAI index.

Sargassum automatically using satellite R_{TOA} signal, and could be applicable to multiple high-resolution satellite sensors.

To better evaluate the performance of the R_{TOA}-RF model, the cross-comparison of *Sargassum* detection results from the R_{TOA}-RF model and the VBFAH method was conducted using the GF1-WFV image on 18 May 2017, as shown in Fig. 7. Here, the GF1-WFV VBFAH product was obtained from satellite *Ref* data [Fig. 7(e)], and the *Sargassum* pixels were then detected with the VBFAH threshold of −0.011 [Fig. 7(f)]. Through interactive visualization of the full image, the *Sargassum* patches extracted by the R_{TOA}-RF model [Fig. 7(b)] were nicely matched with their corresponding patches in RGB image [Fig. 7(a)]. It was also observed that the *Sargassum*result detected by our model was in good consistency with the VBFAH-detected result [see green color in Fig. 7(c)]. The slight differences were mainly found at the edge of macroalgae patches, as shown in Fig. 7(c). In addition, the error matrix of the detection results by using the R_{TOA}-RF model [Fig. 7(b)] and the *Ref*-VBFAH index [Fig. 7(f)] as a "true" result for quantitative assessment, although the VBFAH-detected result was not an absolute truth. The obtaining statistical results for the entire satellite image showed the R_{TOA}-RF model had high accuracy, with the OA, *Kappa*, and $F1$-score values of 99.60%, 88.00%, and 94.00%, respectively. Additionally, the specific details of the good performance of our model were illustrated in the two selected subregions (see bottom row of Fig. 7).

Additionally, the cross-index comparison was made to further evaluate the performance of the R_{TOA}-RF model using the synchronous images of GF6-WFV and LC08-OLI images collected on March 16, 2019 (Fig. 8). In this analysis, the LC08-OLI *Ref*-based FAI data were produced and could show clearly the presence of *Sargassum* features [Fig. 8(e)]. The LC08-OLI *Sargassum* results were detected by the FAI index with its threshold of −0.007 [Fig. 8(f)]. It was observed that the GF6-WFV *Sargassum* patches that were detected by the R_{TOA}-RF model [Fig. 8(b)] matched with the

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visually-identified patches in the RGB image [Fig. 8(a)] and the LC08-OLI FAI-detected *Sargassum* patches [Fig. 8(f)]; moreover, we made the comparison between the result of the R_{TOA}-RF model and the "true" result by manual identification for the selecting sample dataset with 8489 *Sargassum* pixels and 7196 seawater pixels shown in Fig. 8(a). Their error matrix described in Fig. 8(c) showed that the R_{TOA}-RF model can effectively extract the *Sargassum* features with the OA and *Kappa* values of 95.89% and 91.79%. Overall, the results of Figs. 7 and 8 suggested that the R_{TOA}-RF model had high accuracy in detecting *Sargassum* blooms from high-resolution satellite images.

C. Sensitivity Analysis of the R_{TOA}-RF Model

Satellite remote sensing observations often suffer from the changing environmental and observing conditions, such as ocean background (turbid and clear waters), cloud cover, sunglint, and wave glitter. Is the R_{TOA}-RF model robust to these common conditions? To answer this question, the following cases of different high-resolution sensors as illustrative examples were conducted.

Here, two typical conditions with turbid coastal water and clear water were considered for assessing the performance of the R_{TOA}-RF model under different ocean backgrounds, based on two examples of GF1-WFV image over the YECS [Fig. 9(a)–(d)]. For turbid and clear water conditions, no apparent non-*Sargassum* pixels were misclassified as *Sargassum* class by visual inspection; moreover, the R_{TOA}-RF model accurately extracted most *Sargassum* slicks [Fig. 9(b) and (d)] that were visible in the RGB images [Fig. 9(a) and (c)]; however, some low-density *Sargassum* pixels were missed especially under turbid water. In addition, cloud cover and sunglint contamination are common phenomena in high-resolution satellite imagery, adding challenges in object extraction when using optical-based approaches. As an example of *Sargassum* blooms in the YECS, the performance analysis of the R_{TOA}-RF model was investigated under cloud cover and sunglint conditions, as shown in Fig. 9(e)–(h). It was seen that the *Sargassum* pixels were effectively identified under cloud cover and sunglint conditions. Note that cloud pixels were not completely recognized by the proposed model, but fortunately not wrongly classified as *Sargassum*. Certainly, cloud and sunglint masking before model implementation can be conducted to further ensure model accuracy. The above results suggested that this model was less sensitive to observing conditions of background ocean, cloud cover, and sunglint.

Compared to cloud cover and sunglint, surface wave glitter and weak *Sargassum* feature are more challenging for the model performance. Here, more full scenes of satellite images with two challenging conditions were used to justify the model performance, as shown in Figs. 10 and 11. In the analysis, the identification result by the *Ref*-based VBFAH index was treated as a "true result," and then used to evaluate the R_{TOA}-RF model by using the statistical indicators, including OA, *Kappa*, and false negative rate (FNR). For surface wave glitter condition, we found in two GF2-WFV data that the *Sargassum* pixels were detected accurately from wave glitter

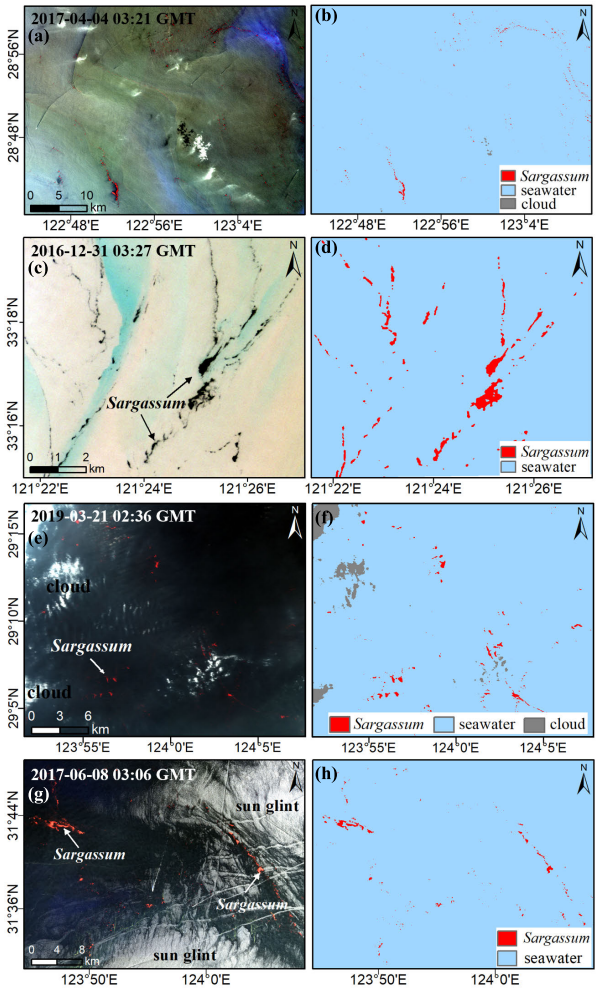


Fig. 9. Satellites images and *Sargassum* results detected by the R_{TOA}-RF model under (a) and (b) turbid water, (c) and (d) clear water, (e) and (f) cloud cover, and (g) and (h) sunglint.

pixels by using the R_{TOA}-RF model, although visual comparison of satellite RGB images [Fig. 10(a) and (e)] and the model results [Fig. 10(b) and (f)]. The *Sargassum* results of the R_{TOA}-RF model matched well with the VBFAH-detected results, with high OA values (>99.60%) and *Kappa* values (>83.00%). Meanwhile, the "noise" pixels (i.e., seawater and wave glitter pixels) were not misidentified as *Sargassum* pixels with low FNR values (<9.60%). Specifically, the high-accuracy detection of *Sargassum* under wave glitter using the R_{TOA}-RF model was comforted by different subregions [see Fig. 10(d) and (h)]. For weak *Sargassum* features condition, it can be qualitatively seen from Fig. 11(a) and (b) that the R_{TOA}-RF model can accurately identify the overwhelming majority of *Sargassum* pixels, including strong and weak *Sargassum*. Especially we clearly found in the selected subregions that most weak *Sargassum* features were effectively extracted [see white circles in Fig. 11(c) for example]; however, some weak *Sargassum* pixels may not be extracted. This was partially motivated by the fact that weak *Sargassum* features are easily confused with water signals (see Fig. 4), and the clear delineation between them is difficult even by manual interpretation [14]. Overall, the results of Figs. 10 and 11 indicated the

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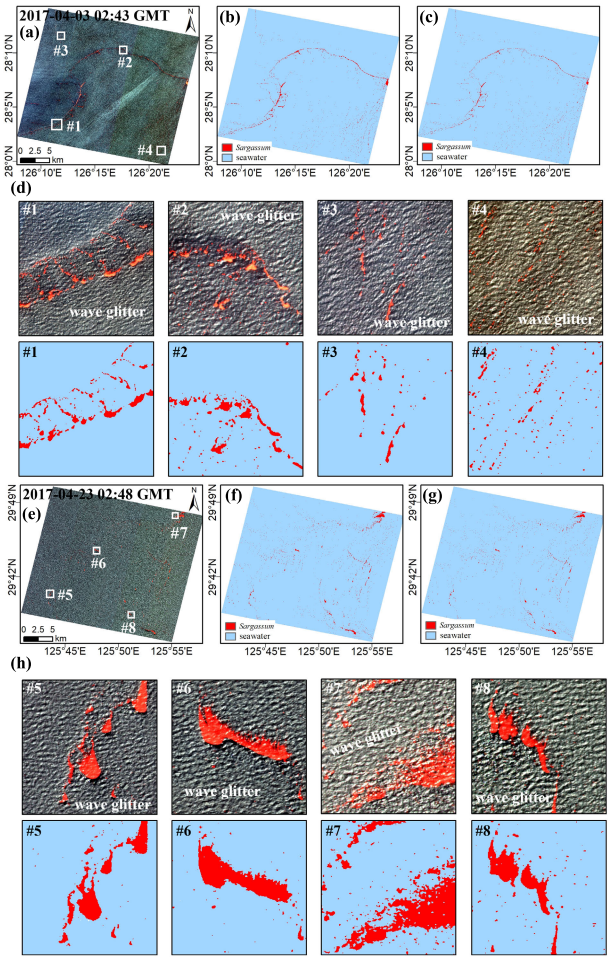


Fig. 10. Performance of the R_{TOA}-RF model under surface wave glitters. (a) and (e) Satellite GF2-MSS RGB images, (b) and (f) their *Sargassum* detection results by using the R_{TOA}-RF model, (c) and (g) their *Sargassum* detection results by using the *Ref*-based VBFAH index, and (d) and (h) detailed performance of the R_{TOA}-RF model in the selected subregions.

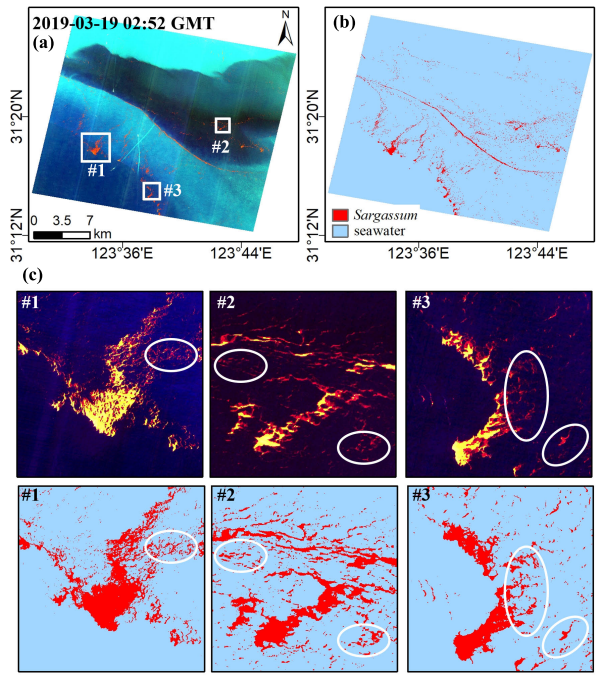


Fig. 11. Performance of the R_{TOA}-RF model under weak *Sargassum* feature. (a) Satellite GF2-MSS RGB image. (b) *Sargassum*result detected by the R_{TOA}-RF model. (c) Detailed performance of the R_{TOA}-RF model in the selected subregions.

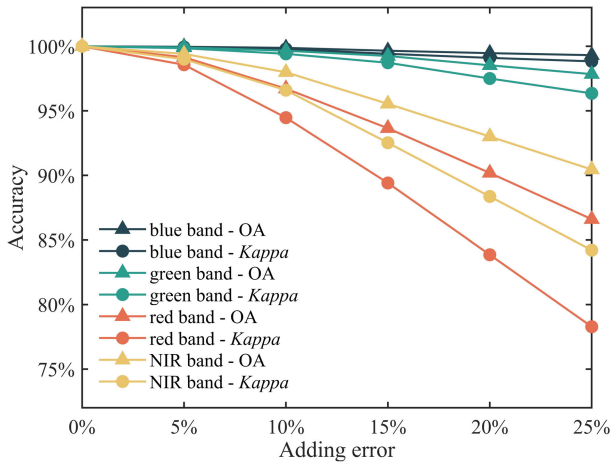


Fig. 12. OA and *Kappa* values as an indication of the model performance with adding random error in the simulated experiments.

R_{TOA}-RF model still had good performance under conditions of wave glitters and weak *Sargassum* features.

To further discuss the model sensitivity to complex observing conditions, the simulated spectrum experiments were carried out here based on the GF1-WFV sample dataset shown in Table III. In the experiments, the random errors with uniform distribution from 0 to 25% in 5% steps were added to the $R_{\mathrm{TOA}}(\lambda)$ values for creating a series of new $R_{\mathrm{TOA}}(\lambda)$ datasets. Then, the R_{TOA}-RF model was applied to each of the new-created R_{TOA} datasets, and the OA and *Kappa* values were calculated for the corresponding error matrix. These processes were repeated 50 times and implemented at blue, green, red, and NIR wavebands, respectively. Fig. 12 displayed the mean accuracy results of 50 repetitions with changes in adding random error. It was found that the OA and *Kappa* values remained relatively high with greater than 95.00% even when the adding random error increased by 25% at blue and green wavebands (see blue and green lines in Fig. 12). These results indicated that the *Sargassum* extraction accuracy of the R_{TOA}-RF model was less affected by the R_{TOA} variations at blue and green wavebands. For the red and NIR wavebands, the OA and *Kappa* values changed dramatically with different adding R_{TOA} errors (see red and beige lines in Fig. 12). Specifically, the model performances remained relatively stable (the OA and *Kappa* values >95.00%) within 10% adding error, and still keep acceptable level (the OA and *Kappa* values > about 85.00%) for the case scenarios within 25% adding error. Additionally, it can be inferred from the comparison of the experimental results that the R_{TOA}-RF model was more sensitive to the R_{TOA} variations at red and NIR wavebands than those at blue and green wavebands. It was largely motivated by the fact that floating *Sargassum* had obvious spectral signatures at red and NIR wavelengths, as shown in Fig. 4. Clearly, the result of the simulated experiments suggested that the R_{TOA}-RF model was a stable method to detect and map floating *Sargassum* blooms from high-resolution satellite data.

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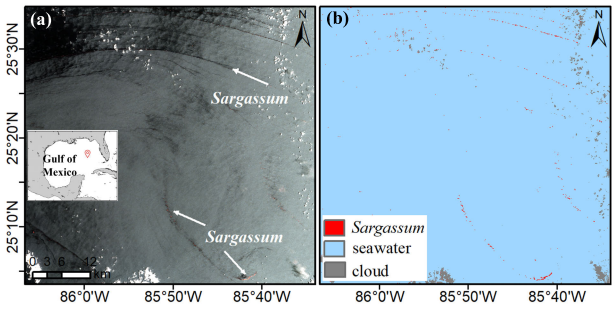


Fig. 13. Example of floating *Sargassum* macroalgae from GF1-WFV scene in the Gulf of Mexico. (a) Satellite RGB image and its location annotated. (b) *Sargassum*result detected by the R_{TOA}-RF model.

D. Further Extension to Detect Sargassum Bloom in the Gulf of Mexico

The above results showed the R_{TOA}-RF model for *Sargassum* detection was implemented to process various high-resolution satellite images for several chosen regions. Specifically, satellite R_{TOA} data and R_{TOA}-based model input features, as defined in Table II, were successively generated. The R_{TOA}-RF model with input features was then applied to extract and map *Sargassum* from satellite imagery. In theory, the processing of the R_{TOA}-RF model could be applied to the global dataset, meaning that it could detect *Sargassum* bloom in the global ocean. Here, taking *Sargassum* bloom in the Gulf of Mexico as an example, we investigated the practical applicability of the R_{TOA}-RF model in the other ocean except for the YECS. Fig. 13 shows an example of the GF1-WFV scene on July 21, 2019, over the Gulf of Mexico, as well as the *Sargassum* mapping detected by the R_{TOA}-RF model. Similar R_{TOA} spectral signatures of *Sargassum* and seawater shown in Fig. 4 were observed in the Gulf of Mexico (not shown here). The detected *Sargassum* patches shown in Fig. 13(b) agreed well with their visual identification in Fig. 13(a), but few low-density *Sargassum* patches were omitted; nevertheless, the example here confirmed the effectiveness of the R_{TOA}-RF model with high-resolution satellite images in detecting *Sargassum* blooms in other parts of the global ocean.

IV. Discussion

A. Comparison of the Model Performances With Different Input Signals

Satellite R_{rc} and *Ref* signals were used to design methods for *Sargassum* detection in previous studies and theoretically could also be used in the RF model. To compare the performance of the R_{TOA}-based RF model with the R_{rc}(*/Ref*)-based RF models, the detailed tests were conducted under different observing conditions. Here for all satellite images used for building the R_{TOA}-RF model, their R_{rc} and *Ref* data were obtained by following the method of Hu et al. [33] and the FLAASH atmospheric correction model, respectively. The sample datasets of R_{rc} and *Ref* with the same pixels shown in Table III were collected from the respective satellite products and were then used to train the RF model with the same input features listed in Table II. For the test satellite images, the *Sargassum* pixels were extracted separately using

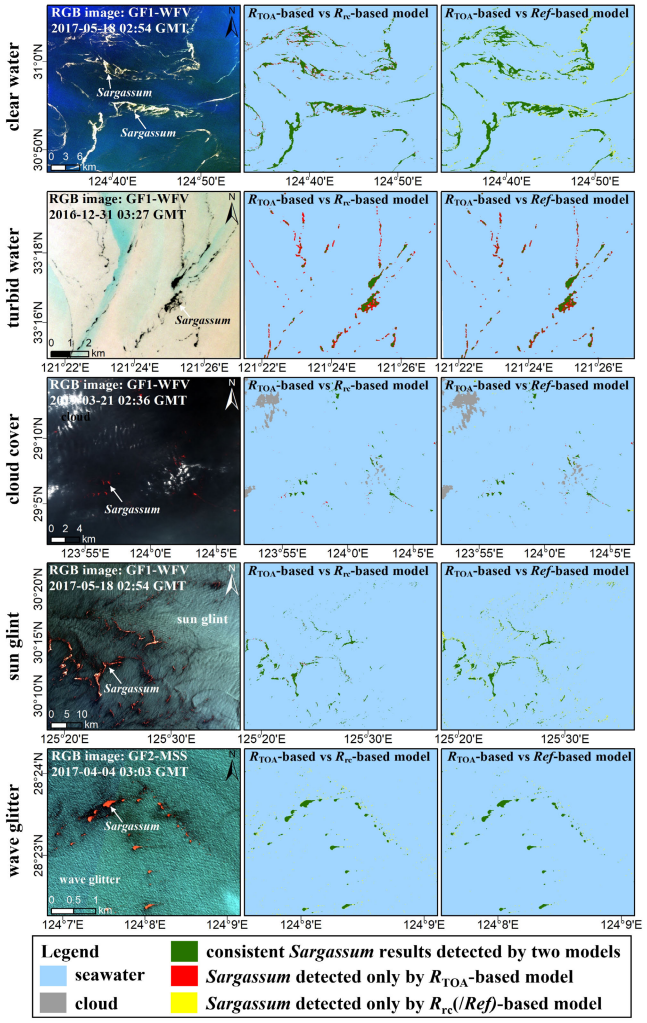


Fig. 14. Comparison of *Sargassum* detection results using the R_{TOA}-based model with (Middle panel) R_{rc}-based model, (Right panel) *Ref*-based model, and (Left panel) Satellite RGB images.

the three trained models, and these results were compared under different observing conditions (Fig. 14).

By visual comparison with the RGB images, it was seen from Fig. 14 that the overwhelming majority of *Sargassum* pixels were successfully extracted by the models trained for three different signals (R_{TOA}, R_{rc}, and *Ref*) under different observing conditions. The performance of the R_{rc}-based model was similar to the R_{TOA}-based model (middle panel of Fig. 14). Indeed, the detection performances of the *Ref*-based model were slightly better than the R_{TOA}-based model, mainly for low-density *Sargassum* slicks as shown in the right panel of Fig. 14. It was because surface reflectance signal manifests the own spectral features of the objects, compared to satellite R_{TOA} data with "noise" signal of atmosphere, thereby improving the model performance to a certain extent. Wave glitter could, however, change the satellite spectral magnitude and comes with the rise of the red-NIR spectral slope, which is reflected obviously in satellite *Ref* and R_{rc} signals. In this event, the pixels with heavy wave glitter may be misclassified as *Sargassum* pixels with satellite R_{rc} and *Ref* data (see bottom row of Fig. 14). The complex processes of

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Rayleigh scattering and atmospheric correction with auxiliary information were required to obtain satellite R_{rc} and *Ref* data, while these processes can be circumvented when generating satellite R_{TOA} data. Thus, the use of satellite R_{TOA} signal for *Sargassum* detection was one advantage of the proposed method.

In summary, the R_{TOA}-RF model can achieve satisfactory performance of *Sargassum* detection under complex observing conditions, similar to the models based on R_{rc} and *Ref*. More importantly, this work proved the feasibility and effectiveness of using satellite R_{TOA} to detect *Sargassum*bloom, improving the data processing speed and efficiency to meet operational monitoring in particular.

B. Advantages and Disadvantages of the R_{TOA}-RF Model

Satellite remote sensing offers unique advantages for providing timely information on *Sargassum* locations and biomasses, helping resource managers to make mitigation strategies [4], [14]. Because of the own detection limit for each sensor [13], high-resolution satellite imagery is more desirable to monitor low-density *Sargassum* patches than coarse-resolution imagery, as clearly evidenced in Fig. 1, and can provide more accurate estimates of *Sargassum* blooms. Based on several high-resolution sensors, a satellite R_{TOA} signal was used to achieve the goal of *Sargassum* detection in this work. The successful use of satellite R_{TOA} in *Sargassum* detection was motived by the different spectral traits of *Sargassum* and background objects shown in Fig. 4. The nine specific R_{TOA}-based features as input of the proposed model were designed for enhancing the spectral differences and achieved satisfactory performances of *Sargassum* detection on various high-resolution satellite sensors (Fig. 6). Note that the scheme of model input features shown in Table III was not fixed and may not be optimal. The main reflectance characteristics of *Sargassum* at red and NIR wavelengths are similar to the green tide of *Ulva* [17], which may obstruct the model accuracy under concurrent green tide and golden tide. It remains to be studied on how to use high-resolution satellite images to distinguish *Sargassum* and *Ulva* as the common floating macroalgae blooms in the coastal waters of China.

This work focused on high-resolution remote detection of *Sargassum* blooms using a satellite R_{TOA} signal with an RF model. Compared to threshold-based segmentation, the current R_{TOA}-RF model required no threshold in extracting *Sargassum* patches, thereby reducing the potential biases caused by the threshold selection [15]. This model showed robust performance with large background variations and various confusing "noise" from cloud, wave glitter, and sunglint (see Figs. 9–11); moreover, the satellite R_{TOA} signal used for *Sargassum* detection makes it easier and faster to interpret vast amounts of satellite images. This is because satellite R_{TOA} products are obtained easily only via simple mathematical calculations. The application flexibility is another advantage of the proposed R_{TOA}-RF model. This model can be successfully applied to different high-resolution satellite sensors, as demonstrated in Fig. 6. Likewise, when the spectral patterns for other high-resolution satellite images were provided, the trained model with respective optimized parameters would achieve satisfactory results of *Sargassum* detection; however, the model performance strongly depends on the training data to be representative of actual classes [34]. In addition, the output results are difficult to diagnose, which is the same for automatic classification methods such as the deep learning model [15]; nevertheless, with a carefully selected training dataset and optimal model parameters, the R_{TOA}-RF model is expected to greatly promote the use of multisource high-resolution satellite images in *Sargassum* monitoring, and is thus, more conducive to disaster management and emergency response.

V. Conclusion

By using satellite R_{TOA} data, an RF-based model, namely the R_{TOA}-RF model, was designed to automatically detect floating *Sargassum* bloom from various high-resolution satellite images. This model was easy to implement with fast processing speeds because of using satellite R_{TOA} data and thereby bypassing atmospheric correction. The examples of the model application on various sensors (GF1-WFV, GF2-MSS, GF6-WFV, HJ1A/B-CCD, and HY1C/D-CZI) achieved high detection accuracy. The sensitivity analysis indicated the R_{TOA}-RF model had satisfactory performances even with the complex ocean background and variable observing "noise" such as cloud cover, sunglint, and surface wave glitters. The R_{TOA}-RF model can also successfully detect *Sargassum* bloom in the Gulf of Mexico. The findings of this work support to the practicability of satellite R_{TOA} signal and the potential of high-resolution satellite imagery for detecting *Sargassum* in the global ocean. The concurrent floating macroalgae blooms may increase the uncertainty of *Sargassum* detection. Future efforts are thus expected to identify mixed macroalgae from high-resolution satellite imagery to promote the fine-scale monitoring of marine algae blooms.

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Hailong Zhang received the master's degree in geomatics engineering and the Ph.D. degree in marine meteorology from the Nanjing University of Information Science and Technology (NUIST), Nanjing, China, in 2015 and 2019, respectively.

He has been working with the Marine Environmental Parameter Inversion and Marine Ecological Disaster Monitoring since 2015.



Jianqiang Liu received the B.S. degree in geophysics from Peking University, Beijing, China, in 1986, and the master's degree in marine meteorology from the National Marine Environmental Forecasting Center, State Oceanic Administration, Beijing, in 1989.

He is the Chief Designer of the ground system for the Chinese-France Ocean satellite and new generational ocean color satellites and was a Deputy Chief Designer of the ground application system for the HY-1 and HY-2 satellites. He is one of the Founders of satellite ocean remote sensing in China and plays an important role in the development of Chinese ocean satellites and manned space flight. His research interests include ocean color and sea wave satellite data processing, applications of ocean remote sensing data, and Antarctic research.



Xiaomin Ye received the B.S. degree in optical information science and technology from Nankai University, Tianjin, China, in 2006, the master's degree in physical oceanography from the First Institute of Oceanography, State Oceanic Administration, Qingdao, China, in 2009, and the Ph.D. degree in oceanic detection technology from the Ocean University of China, Qingdao, in 2017.

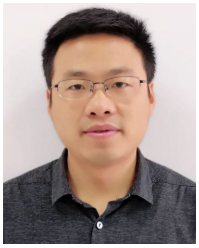
He is currently the Deputy Chief Designer of the ground system of the Chinese geostationary ocean and coastal zone environment satellite and a Designer-in-Charge of data processing for the Chinese ocean color satellites, National Satellite Ocean Application Service, Beijing, China. His research interests include electromagnetic wave scattering from the ocean surface and the application of ocean satellite remote sensing data.

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Yan Bai received the B.S. and M.S. degrees from Sun Yat-sen University, Guangzhou, China, in 2000 and 2003, respectively, and the Ph.D. degree in electromagnetic fields and microwave techniques from the Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai, China, in 2007.

She is currently with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, MNR, Hangzhou, China, the Southern Marine Science and Engineering Guangdong Laboratory, Guangzhou, and the School of Oceanography, Shanghai Jiao Tong University, Shanghai. Her research interests are satellite remote sensing of the marine carbon cycle.



Deyong Sun received the B.S. degree in geomatics engineering from Nanjing Tech University, Nanjing, China, in 2009, and the Ph.D. degree in geographical science from Nanjing Normal University, Nanjing, in 2013.

From 2018, he has been working in the coastal water optical properties, environmental remote sensing, coastal suspended particle size, phytoplankton size classes and evolvement rules, and sudden marine disaster phenomenon fields at the Nanjing University of Information Science and Technology, Nanjing.



Shengqiang Wang received the B.S. degree in physics from Beijing Normal University, Beijing, China, in 2008, and the Ph.D. degree in environmentology from Nagoya University, Nagoya, Japan, in 2014.

Since 2015, he has been working in the ocean optical remote sensing, marine bio-optics, and satellite oceanography fields at the Nanjing University of Information Science and Technology, Nanjing, China.



Jingming Dong received the B.S. degree in remote sensing science and technology from the Binjiang College, Nanjing, China, in 2012, and the M.S. degree in applied meteorology from the Nanjing University of Information Science and Technology, Nanjing, in 2015.

He has worked in marine meteorology and application of meteorological satellites.

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